



Voice Spectrum Analyzer

A real-time voice recording and spectral analysis tool built in **GNU Octave**, developed as part of the ECE231 Signals & Systems course at **Nile University**.



Team Members

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Course: ECE231 – Signals & Systems

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Project Overview

This project implements a full-featured Voice Spectrum Analyzer that records a human voice signal via microphone and performs both time-domain and frequency-domain analysis. An interactive GUI allows full control over recording parameters, visualization, and file saving.



Features

-  Real-time audio recording with live waveform preview
-  Time-domain signal visualization with dynamic axis scaling
-  Frequency-domain analysis using FFT (Amplitude, Phase, and Energy spectra)

- 🎧 Automatic silence detection based on RMS amplitude threshold and duration
 - 🖥️ Interactive GUI built with Octave `uicontrol` and `axes`
 - 💾 Save recorded audio as `.wav` and plots as image files
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System Flowchart

```
Start → Initialize GUI → User Input (Fs, Tmax, Thresholds)
      → Start Recording → Real-time Buffering
      → Silence Detection? → Yes: Stop | No: Continue
      → Time-Domain Analysis → FFT Computation
      → Display Plots (Amplitude / Phase / Energy)
      → Save Audio & Plots → End
```

Implementation Details

GUI Parameters

- Maximum recording time
- Sampling frequency (default: 8000 Hz)
- Silence amplitude threshold (default: 0.0001)
- Silence duration threshold (default: 1 s)
- Save folder selection

Signal Processing Pipeline

1. Audio captured via `audiorecorder`
 2. Signal divided into short frames for RMS-based silence detection
 3. FFT computed on full recorded signal
 4. Positive frequency components extracted with proper normalization
 5. Amplitude, phase, and energy spectra computed and displayed
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Results Summary

Analysis	Observation
Time Domain	Clear waveform with visible silent segments
Amplitude Spectrum	Main voice energy concentrated below 4 kHz
Phase Spectrum	Phase values across $\pm\pi$ range
Energy Spectrum	Dominant low-frequency components
Silence Detection	Accurate stop without false triggers on brief pauses

Contribution Table

Member	Functions	Role
Ali Mohamed Ali	<code>voice_analyzer_full , startRecordingCB</code>	Main GUI layout & audio recording
Ahmed Gamal	<code>createOutputWindows_Final , updateFreqLayout_fromPopup</code>	Output windows & plot switching
Omar Gamal	<code>applyFreqZoom , applyTimeZoom</code>	Axis scaling & zoom control
Ziad Bedir	<code>stopRecordingCB , recordAgainCB</code>	Recording control & reset
Amr Mahmoud	<code>SaveAudioCB , savePlotsCB , browseFolder</code>	File saving & folder selection

How to Run

1. Open **GNU Octave**
2. Navigate to the project directory
3. Run the main script:

voice_spectrum_analyzer

4. Set your parameters in the GUI and click **Start Recording**

File Structure

```
voice_spectrum_analyzer/  
|  
├─ voice_spectrum_analyzer.m    # Main source file (all functions included)  
└─ README.md
```

Note: The full source code is contained in a single modular `.m` file and is also available on Moodle.